

3G Communication System: Complete Source Code

File: 3G_COMMUNICATION_SYSTEM_STANDALONE.html • Self-contained Single-File Executable Application (React 18, Tailwind CSS, Canvas Simulation, State Machine, QoS Queues, Soft Handover, and Capacity Calculator).

```

1 | <!DOCTYPE html>
2 | <html lang="en" class="dark">
3 | <head>
4 |   <meta charset="UTF-8">
5 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 |   <title>3G Real-Time Communication System | Mobile Video Surveillance & Emergency Monitoring</title>
7 |   <meta name="description" content="Complete standalone interactive simulation of a 3G communication system for real-time mobile video surveillance, emergency monitoring, and network management.">
8 |   <!-- Fonts -->
9 |   <link rel="preconnect" href="https://fonts.googleapis.com">
10 |   <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
11 |   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&family=JetBrainsMono:wght@400;500;600;700" rel="stylesheet">
12 |   <!-- Tailwind CSS via CDN -->
13 |   <script src="https://cdn.tailwindcss.com"></script>
14 |   <!-- React 18 & ReactDOM via CDN -->
15 |   <script src="https://unpkg.com/react@18/umd/react.production.min.js" crossorigin></script>
16 |   <script src="https://unpkg.com/react-dom@18/umd/react-dom.production.min.js" crossorigin></script>
17 |   <!-- Babel for inline JSX -->
18 |   <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
19 |
20 |   <style>
21 |     :root {
22 |       --font-sans: 'Inter', system-ui, sans-serif;
23 |       --font-mono: 'JetBrains Mono', monospace;
24 |       --font-display: 'Outfit', sans-serif;
25 |     }
26 |     body {
27 |       font-family: var(--font-sans);
28 |       background-color: #030712;
29 |       color: #f3f4f6;
30 |       overflow-x: hidden;
31 |       margin: 0;
32 |       padding: 0;
33 |     }
34 |     .telecom-grid {
35 |       background-size: 32px 32px;
36 |       background-image:
37 |         linear-gradient(to right, rgba(255, 255, 255, 0.04) 1px, transparent 1px),
38 |         linear-gradient(to bottom, rgba(255, 255, 255, 0.04) 1px, transparent 1px);
39 |     }
40 |     .surveillance-scanlines {
41 |       background: linear-gradient(rgba(18, 16, 16, 0) 50%, rgba(0, 0, 0, 0.35) 50%),
42 |         linear-gradient(90deg, rgba(255, 0, 0, 0.03), rgba(0, 255, 0, 0.01), rgba(0, 0, 255, 0.03));
43 |       background-size: 100% 3px, 4px 100%;
44 |       pointer-events: none;
45 |     }
46 |     .glow-cyan {
47 |       box-shadow: 0 0 20px -3px rgba(6, 182, 212, 0.35);
48 |     }
49 |     .glow-red {
50 |       box-shadow: 0 0 25px 2px rgba(239, 68, 68, 0.5);
51 |       animation: pulse-red 1.2s cubic-bezier(0.4, 0, 0.6, 1) infinite;
52 |     }
53 |     @keyframes pulse-red {
54 |       0%, 100% { opacity: 1; box-shadow: 0 0 25px 2px rgba(239, 68, 68, 0.6); }
55 |       50% { opacity: 0.85; box-shadow: 0 0 10px 0px rgba(239, 68, 68, 0.2); }
56 |     }
57 |     ::-webkit-scrollbar { width: 6px; height: 6px; }
58 |     ::-webkit-scrollbar-track { background: #0b0f19; }
59 |     ::-webkit-scrollbar-thumb { background: #1e293b; border-radius: 3px; }
60 |     ::-webkit-scrollbar-thumb:hover { background: #334155; }
61 |   </style>
62 | </head>

```

```

63 | <body class="bg-slate-950 text-slate-100 antialiased selection:bg-cyan-500/30 selection:text-cyan-200">
64 |   <div id="root"></div>
65 |
66 |   <!-- Inline Application Script using Babel Standalone -->
67 |   <script type="text/babel">
68 |     const { useState, useEffect, useRef, useCallback } = React;
69 |
70 |     // SVG Icons Helper
71 |     const Icon = ({ name, className = "w-4 h-4" }) => {
72 |       const icons = {
73 |         radio: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
74 |         video: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
75 |         play: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
76 |         pause: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
77 |         reset: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
78 |         forward: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
79 |         car: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
80 |         alarm: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
81 |         network: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
82 |         sliders: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
83 |         calc: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
84 |         shield: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
85 |         clock: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
86 |         chart: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
87 |         book: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
88 |         check: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
89 |         crosshair: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
90 |         eye: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
91 |         send: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
92 |         close: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLin
93 |       };
94 |       return icons[name] || <span />;
95 |     };
96 |
97 |     // System Static Data
98 |     const SYSTEM_NODES = [
99 |       {
100 |         id: 'camera-unit',
101 |         name: 'Mobile Camera Unit',
102 |         shortName: 'Mobile Camera',
103 |         subtitle: 'Edge Capture & 3G UE',
104 |         category: 'User Equipment',
105 |         purpose: 'Captures raw high-definition video and ambient audio, compresses the media into optimized digital streams, and tra
106 |         functions: [
107 |           'Visual & audio sensor acquisition (Camera sensor & directional mic)',
108 |           'Hardware video encoding (H.264 Baseline Profile / MPEG-4 Part 2)',
109 |           'Packetization into RTP/UDP/IP payloads',
110 |           '3G/WCDMA User Equipment (UE) modem transmission',
111 |           'Power management via internal high-capacity battery'
112 |         ],
113 |         dataHandled: 'Raw video frames, compressed video bitstream (256 kbps baseline), audio packets, telemetry.',
114 |         roleInRealTime: 'Originates delay-sensitive video, audio, and emergency alarm packets. Adjusts encoder rate dynamically.',
115 |         subComponents: ['Optical Lens & CMOS Sensor', 'Omni Microphone', 'Hardware Video Encoder', '3G WCDMA Modem (UE)', 'Li-ion Ba
116 |       },
117 |       {
118 |         id: 'node-b',
119 |         name: 'Node B (Base Station)',
120 |         shortName: 'Node B',
121 |         subtitle: 'WCDMA Radio Access',
122 |         category: 'Radio Access Network',
123 |         purpose: 'Provides physical WCDMA radio layer connectivity between mobile camera units and the fixed terrestrial radio netwo
124 |         functions: [

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125 |         'WCDMA radio frequency transmission and reception (5 MHz carrier)',
126 |         'Spreading and despreading using orthogonal channelization codes (OVSF)',
127 |         'Inner-loop fast power control (1,500 Hz slot rate)',
128 |         'Air interface modulation, demodulation, and channel coding'
129 |     ],
130 |     dataHandled: 'Air interface radio frames, WCDMA chip-level symbols (3.84 Mcps), dedicated physical channels (DPCH/PRACH).',
131 |     roleInRealTime: 'Ensures reliable low-latency wireless transmission over the wireless link and monitors instantaneous channel quality',
132 |     subComponents: ['Radio Transceiver (TRX)', 'Power Amplifiers', 'Duplexer & Antennas', 'Baseband DSP Unit']
133 | },
134 | {
135 |     id: 'rnc',
136 |     name: 'Radio Network Controller (RNC)',
137 |     shortName: 'RNC',
138 |     subtitle: 'Radio Resource Management',
139 |     category: 'Radio Access Network',
140 |     purpose: 'Governs all radio resources across multiple Node Bs, coordinates seamless handovers, and manages admission control',
141 |     functions: [
142 |         'Radio Resource Management (RRM) & dynamic channel allocation',
143 |         'Soft handover and harder handover control between cells',
144 |         'Admission control for new real-time voice and video sessions',
145 |         'Radio Access Bearer (RAB) establishment and QoS mapping'
146 |     ],
147 |     dataHandled: 'RLC protocol data units (PDUs), radio signaling messages (NBAP, RRC, RANAP), user payload frames.',
148 |     roleInRealTime: 'Prioritizes real-time media streams and executes seamless soft handover without dropping surveillance video',
149 |     subComponents: ['Call Control Processor', 'Frame Processing Unit', 'ATM/IP Switch Fabric']
150 | },
151 | {
152 |     id: 'packet-core',
153 |     name: '3G Packet Core (SGSN / GGSN)',
154 |     shortName: '3G Packet Core',
155 |     subtitle: 'Serving & Gateway GPRS Nodes',
156 |     category: 'Core Network',
157 |     purpose: 'Performs packet-switched routing, subscriber authentication, mobility management, and QoS enforcement.',
158 |     functions: [
159 |         'SGSN: Mobility management, location tracking, and logical link control',
160 |         'GGSN: Gateway to external IP networks, IP address allocation (DHCP/Radius)',
161 |         '3GPP AKA mutual subscriber authentication and cryptographic ciphering',
162 |         'Packet Data Protocol (PDP) context activation & management'
163 |     ],
164 |     dataHandled: 'GTP-U (GPRS Tunneling Protocol User plane) tunnels, IP packets, charging records (CDRs).',
165 |     roleInRealTime: 'Enforces end-to-end QoS and maintains IP address persistence during mobility.',
166 |     subComponents: ['SGSN (Serving GPRS Support Node)', 'GGSN (Gateway GPRS Support Node)', 'HLR/AuC Connector']
167 | },
168 | {
169 |     id: 'private-ip',
170 |     name: 'Private IP Network / Internet',
171 |     shortName: 'Private IP Network',
172 |     subtitle: 'Secure Core Transport',
173 |     category: 'Transport',
174 |     purpose: 'Transports routed IP packets securely across private leased lines or encrypted VPN tunnels to the application servers',
175 |     functions: [
176 |         'IP packet routing and high-speed multi-gigabit switching',
177 |         'IPsec VPN tunneling for end-to-end data confidentiality',
178 |         'DiffServ / MPLS Traffic Engineering (QoS preservation)'
179 |     ],
180 |     dataHandled: 'Encapsulated IPsec ESP packets, RTP/RTCP surveillance video streams, RTSP messages.',
181 |     roleInRealTime: 'Maintains bounded latency and low packet loss across the wide area network.',
182 |     subComponents: ['Edge Routers', 'IPsec VPN Concentrator', 'Next-Gen Firewall']
183 | },
184 | {
185 |     id: 'app-server',
186 |     name: 'Application / Streaming Server',
```

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187 |     shortName: 'Streaming Server',
188 |     subtitle: 'Session & Media Gateway',
189 |     category: 'Application',
190 |     purpose: 'Terminates camera streaming sessions, distributes live feeds to authorized monitoring terminals, and records footage',
191 |     functions: [
192 |       'RTSP/SIP session initiation, teardown, and heartbeat monitoring',
193 |       'RTP media packet ingestion and jitter buffer smoothing',
194 |       'Stream fan-out and re-transmission to multiple operator consoles',
195 |       'Emergency alarm processing, notification dispatch, and event logging'
196 |     ],
197 |     dataHandled: 'Inbound RTP streams, outbound replicated unicast streams, RTSP commands, alarm event webhooks.',
198 |     roleInRealTime: 'Central relay point for real-time monitoring; detects packet drop and triggers rapid rate adaptation.',
199 |     subComponents: ['RTSP/RTMP Media Server', 'Session Controller', 'Event Dispatch Engine', 'NVR Storage']
200 |   },
201 |   {
202 |     id: 'monitoring-terminal',
203 |     name: 'Monitoring Terminal / Center',
204 |     shortName: 'Monitoring Console',
205 |     subtitle: 'Operator Display & PTZ Control',
206 |     category: 'Terminal',
207 |     purpose: 'Decodes compressed video in real time, displays live surveillance feeds to control-room operators, and facilitates coordination',
208 |     functions: [
209 |       'Hardware-accelerated H.264/MPEG-4 video decoding',
210 |       'Low-latency multi-channel surveillance matrix display',
211 |       'Audio playback and two-way voice channel to mobile unit',
212 |       'Camera PTZ control commands transmission',
213 |       'Emergency alert audio-visual popups and operator dispatch interface'
214 |     ],
215 |     dataHandled: 'Decoded video frames, audio samples, operator control commands, real-time diagnostic telemetry.',
216 |     roleInRealTime: 'Final consumer of the surveillance stream and origin of operator commands. Requires low latency (< 350 ms).',
217 |     subComponents: ['Workstation GPU Decoder', 'Ultra-Wide Operator Displays', 'PTZ Joystick Controller']
218 |   }
219 | ];
220 |
221 | const SCENARIO_STEPS = [
222 |   { step: 1, title: 'Camera Boot & 3G UE Registration', state: 'CONNECTED', desc: 'The mobile camera powers on. Its 3G modem registers with the network.' },
223 |   { step: 2, title: 'Video Stream Start & QoS Allocation', state: 'STREAMING', desc: 'Camera initiates RTSP session. RNC allocates resources for streaming.' },
224 |   { step: 3, title: 'Normal Mobile Patrol & Link Maintenance', state: 'MOVING', desc: 'Patrol vehicle moves through Cell A. Fast handover occurs to Cell B.' },
225 |   { step: 4, title: 'Cell Boundary Reached & Seamless Handover', state: 'HANDOVER', desc: 'Vehicle approaches cell edge. RNC initiates handover to Cell C.' },
226 |   { step: 5, title: 'Network Congestion & QoS Traffic Protection', state: 'CONGESTED', desc: 'Heavy background traffic occurs. Scheduling prioritizes emergency feeds.' },
227 |   { step: 6, title: 'Emergency Alarm Trigger & Immediate Dispatch', state: 'EMERGENCY', desc: 'Tactical emergency tripped. High-priority alert is dispatched to all consoles.' },
228 |   { step: 7, title: 'Session Teardown & Secure Audit Archiving', state: 'SESSION_END', desc: 'Mission concludes. Radio bearer is released. Session logs are securely archived.' }
229 | ];
230 |
231 | // Main App Component
232 | function App() {
233 |   const [activeTab, setActiveTab] = useState('dashboard');
234 |   const [systemState, setSystemState] = useState('IDLE');
235 |   const [isTransmitting, setIsTransmitting] = useState(false);
236 |   const [isMobilityRunning, setIsMobilityRunning] = useState(false);
237 |   const [selectedNode, setSelectedNode] = useState(null);
238 |   const [activeNodePulse, setActiveNodePulse] = useState(null);
239 |   const [packets, setPackets] = useState([]);
240 |   const [activeCamId, setActiveCamId] = useState('CAM-01');
241 |   const [visionMode, setVisionMode] = useState('NORMAL');
242 |   const [showCrosshairs, setShowCrosshairs] = useState(true);
243 |   const [showMotionBox, setShowMotionBox] = useState(true);
244 |   const [emergencyAlert, setEmergencyAlert] = useState(null);
245 |
246 |   // Congestion & Telemetry State
247 |   const [networkLoad, setNetworkLoad] = useState('LOW');
248 |   const [signalDbm, setSignalDbm] = useState(-80);

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249 |     const [activeCameras, setActiveCameras] = useState(1);
250 |     const [cameraBitrate, setCameraBitrate] = useState(256);
251 |     const [admissionStatus, setAdmissionStatus] = useState('NORMAL');
252 |     const [telemetry, setTelemetry] = useState({
253 |       bitrateKbps: 256,
254 |       latencyMs: 142,
255 |       packetLossPct: 0.1,
256 |       packetsSent: 0,
257 |       packetsReceived: 0,
258 |       packetsDropped: 0,
259 |       queueHigh: 0,
260 |       queueLow: 0,
261 |       emergencyCount: 0
262 |     });
263 |
264 |     // Mobility State
265 |     const [handoverState, setHandoverState] = useState({
266 |       activeCell: 'Cell A',
267 |       targetCell: null,
268 |       status: 'IDLE',
269 |       cellASignal: -8.2,
270 |       cellBSignal: -17.5,
271 |       cellCSignal: -22.0,
272 |       distanceKm: 0.8,
273 |       handoversCount: 0
274 |     });
275 |
276 |     // Scenario State
277 |     const [scenarioIndex, setScenarioIndex] = useState(0);
278 |     const [isScenarioRunning, setIsScenarioRunning] = useState(false);
279 |     const [scenarioProgress, setScenarioProgress] = useState(0);
280 |     const [scenarioSpeed, setScenarioSpeed] = useState(1);
281 |
282 |     // Event Logs & Audit Logs
283 |     const [eventLogs, setEventLogs] = useState([
284 |       { id: 1, time: '21:00:00', source: 'SYSTEM', msg: 'System initialized. 3G UMTS network architecture standing by.', type: 'INFO' },
285 |       { id: 2, time: '21:00:01', source: 'NODE-B', msg: 'WCDMA Carrier 2100 MHz active. 3.84 Mcps chip clock synchronized.', type: 'INFO' },
286 |       { id: 3, time: '21:00:02', source: 'RNC', msg: 'Radio Network Controller ready. Radio Bearer pool initialized.', type: 'INFO' }
287 |     ]);
288 |
289 |     const [auditLogs, setAuditLogs] = useState([
290 |       { id: 1, time: '21:00:01', device: 'CAMERA-01', user: 'Operator-01', action: 'System Ready Check', status: 'SUCCESS', details: '' }
291 |     ]);
292 |
293 |     const nodeSeq = ['camera-unit', 'node-b', 'rnc', 'packet-core', 'private-ip', 'app-server', 'monitoring-terminal'];
294 |
295 |     const addLog = (source, msg, type = 'INFO') => {
296 |       const time = new Date().toTimeString().split(' ')[0];
297 |       setEventLogs(prev => [{ id: Date.now() + Math.random(), time, source, msg, type }, ...prev.slice(0, 79)]);
298 |     };
299 |
300 |     const addAudit = (device, user, action, status, details) => {
301 |       const time = new Date().toTimeString().split(' ')[0];
302 |       setAuditLogs(prev => [{ id: Date.now(), time, device, user, action, status, details }, ...prev.slice(0, 39)]);
303 |     };
304 |
305 |     // Admission Control Check
306 |     useEffect(() => {
307 |       if (activeCameras * cameraBitrate > 3840) {
308 |         setAdmissionStatus('REJECTED');
309 |         addLog('RNC-ADMISSION', `Admission Control: Rejected session for Camera #${activeCameras}. Insufficient capacity.`, 'WARNING');
310 |       } else {

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311 |         setAdmissionStatus('NORMAL');
312 |     }
313 | }, [activeCameras, cameraBitrate]);
314 |
315 | // Manual Packet Sender
316 | const sendPacket = (type, isAlarm = false) => {
317 |     const colors = { VIDEO: '#06b6d4', AUDIO: '#a855f7', CONTROL: '#f59e0b', ALARM: '#ef4444', BACKGROUND: '#64748b' };
318 |     const pkt = {
319 |         id: Date.now() + Math.random(),
320 |         type,
321 |         priority: isAlarm || type === 'ALARM' ? 'CRITICAL' : type === 'BACKGROUND' ? 'LOW' : 'HIGH',
322 |         currentLink: 0,
323 |         progress: 0,
324 |         color: colors[type] || '#06b6d4',
325 |         label: type,
326 |         isAlarm
327 |     };
328 |     setPackets(prev => [...prev, pkt]);
329 |     setTelemetry(t => ({
330 |         ...t,
331 |         packetsSent: t.packetsSent + 1,
332 |         queueHigh: pkt.priority !== 'LOW' ? (t.queueHigh + 1) % 15 : t.queueHigh,
333 |         queueLow: pkt.priority === 'LOW' ? (t.queueLow + 1) % 12 : t.queueLow
334 |     }));
335 |     addLog('CAMERA-01', `Manual ${type} packet injected into radio buffer.`, isAlarm ? 'EMERGENCY' : 'INFO');
336 | };
337 |
338 | // Transmission Packet Interval
339 | useEffect(() => {
340 |     if (!isTransmitting) return;
341 |     const interval = setInterval(() => {
342 |         const r = Math.random();
343 |         let type = 'VIDEO';
344 |         if (r > 0.85) type = 'AUDIO';
345 |         else if (r > 0.75) type = 'CONTROL';
346 |         else if (r > 0.65) type = 'BACKGROUND';
347 |
348 |         if (networkLoad === 'CONGESTED' && type === 'BACKGROUND' && Math.random() > 0.3) {
349 |             setTelemetry(t => ({ ...t, packetsDropped: t.packetsDropped + 1 }));
350 |             return;
351 |         }
352 |         sendPacket(type, false);
353 |     }, networkLoad === 'CONGESTED' ? 750 : 500);
354 |
355 |     return () => clearInterval(interval);
356 | }, [isTransmitting, networkLoad]);
357 |
358 | // Packet Physics RAF
359 | useEffect(() => {
360 |     if (packets.length === 0) return;
361 |     const anim = requestAnimationFrame(() => {
362 |         setPackets(prev => {
363 |             const next = [];
364 |             for (const pkt of prev) {
365 |                 let spd = 0.04;
366 |                 if (pkt.isAlarm) spd = 0.07;
367 |                 if (pkt.priority === 'LOW' && networkLoad === 'CONGESTED') spd = 0.015;
368 |                 const prog = pkt.progress + spd;
369 |                 if (prog >= 1) {
370 |                     const nextLink = pkt.currentLink + 1;
371 |                     const hitNode = nodeSeq[nextLink];
372 |                     setActiveNodePulse(hitNode);

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373 |         setTimeout(() => setActiveNodePulse(null), 250);
374 |         if (nextLink >= 6) {
375 |             setTelemetry(t => ({ ...t, packetsReceived: t.packetsReceived + 1 }));
376 |         } else {
377 |             next.push({ ...pkt, currentLink: nextLink, progress: 0 });
378 |         }
379 |     } else {
380 |         next.push({ ...pkt, progress: prog });
381 |     }
382 | }
383 | return next;
384 | });
385 | });
386 | return () => cancelAnimationFrame(anim);
387 | }, [packets, networkLoad]);
388 |
389 | // Mobility & Handover Loop
390 | useEffect(() => {
391 |     if (!isMobilityRunning) return;
392 |     const interval = setInterval(() => {
393 |         setHandoverState(prev => {
394 |             let dist = prev.distanceKm + 0.15;
395 |             if (dist > 9.0) dist = 0.5;
396 |             const aSig = Math.max(-25, -6.5 - Math.pow(dist - 1.5, 2) * 1.5);
397 |             const bSig = Math.max(-25, -6.5 - Math.pow(dist - 4.5, 2) * 1.5);
398 |             const cSig = Math.max(-25, -6.5 - Math.pow(dist - 7.5, 2) * 1.5);
399 |
400 |             let st = prev.status;
401 |             let active = prev.activeCell;
402 |             let target = prev.targetCell;
403 |             let count = prev.handoversCount;
404 |
405 |             if (active === 'Cell A' && dist >= 2.8 && dist <= 3.4) {
406 |                 if (dist >= 3.0 && st !== 'HANDOVER_INITIATED') {
407 |                     st = 'HANDOVER_INITIATED';
408 |                     target = 'Cell B';
409 |                     addLog('RNC-HANDOVER', 'HANDOVER INITIATED: Cell A -> Cell B Active Set Update (Soft Handover)', 'WARNING');
410 |                     setSystemState('HANDOVER');
411 |                 }
412 |             } else if (active === 'Cell A' && dist > 3.4) {
413 |                 st = 'HANDOVER_SUCCESSFUL';
414 |                 active = 'Cell B';
415 |                 target = null;
416 |                 count += 1;
417 |                 addLog('RNC-HANDOVER', 'HANDOVER SUCCESSFUL: Transferred seamlessly to Cell B. 0 frame freeze.', 'SUCCESS');
418 |                 setSystemState(isTransmitting ? 'STREAMING' : 'MOVING');
419 |             }
420 |             return { ...prev, distanceKm: parseFloat(dist.toFixed(2)), cellASignal: parseFloat(aSig.toFixed(1)), cellBSignal: parseF
421 |         });
422 |     }, 600);
423 |     return () => clearInterval(interval);
424 | }, [isMobilityRunning, isTransmitting]);
425 |
426 | // Surveillance Video Canvas Rendering
427 | const canvasRef = useRef(null);
428 | useEffect(() => {
429 |     const canvas = canvasRef.current;
430 |     if (!canvas) return;
431 |     const ctx = canvas.getContext('2d');
432 |     if (!ctx) return;
433 |     let anim;
434 |     let tx = 180, ty = 120, vx = 0.8, vy = 0.4;

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```

435 |
436 |     const render = () => {
437 |         const w = canvas.width, h = canvas.height;
438 |         ctx.fillStyle = visionMode === 'IR_THERMAL' ? '#180728' : '#060c18';
439 |         ctx.fillRect(0, 0, w, h);
440 |
441 |         if (!isTransmitting && systemState !== 'STREAMING' && systemState !== 'MOVING' && systemState !== 'EMERGENCY') {
442 |             ctx.fillStyle = '#0f172a';
443 |             ctx.fillRect(0, 0, w, h);
444 |             ctx.fillStyle = '#94a3b8';
445 |             ctx.font = 'bold 15px "JetBrains Mono", monospace';
446 |             ctx.textAlign = 'center';
447 |             ctx.fillText('CAMERA STANDBY / TRANSMISSION PAUSED', w / 2, h / 2 - 10);
448 |             ctx.font = '12px "Inter", sans-serif';
449 |             ctx.fillStyle = '#64748b';
450 |             ctx.fillText('Click "START LIVE TRANSMISSION" above to establish 3G video uplink', w / 2, h / 2 + 15);
451 |             return;
452 |         }
453 |
454 |         // Horizon & Buildings
455 |         ctx.strokeStyle = visionMode === 'IR_THERMAL' ? '#581c87' : '#1e293b';
456 |         ctx.beginPath(); ctx.moveTo(0, h * 0.65); ctx.lineTo(w, h * 0.65); ctx.stroke();
457 |         ctx.fillStyle = visionMode === 'IR_THERMAL' ? '#3b0764' : '#0a1526';
458 |         ctx.fillRect(50, h * 0.35, 90, h * 0.3); ctx.fillRect(170, h * 0.28, 120, h * 0.37); ctx.fillRect(320, h * 0.4, 110, h * 0.3);
459 |
460 |         // Moving Vehicle
461 |         tx += vx; ty += vy;
462 |         if (tx > w - 100 || tx < 80) vx *= -1;
463 |         if (ty > h * 0.75 || ty < h * 0.55) vy *= -1;
464 |         ctx.fillStyle = visionMode === 'IR_THERMAL' ? '#ef4444' : '#334155';
465 |         ctx.fillRect(tx, ty, 60, 24);
466 |         ctx.fillStyle = visionMode === 'IR_THERMAL' ? '#fbbf24' : '#1e293b';
467 |         ctx.fillRect(tx + 12, ty - 12, 34, 12);
468 |
469 |         // Motion Box
470 |         if (showMotionBox) {
471 |             ctx.strokeStyle = systemState === 'EMERGENCY' ? '#ef4444' : '#22c55e';
472 |             ctx.lineWidth = 1.5;
473 |             ctx.strokeRect(tx - 8, ty - 18, 76, 50);
474 |             ctx.fillStyle = systemState === 'EMERGENCY' ? '#ef4444' : '#22c55e';
475 |             ctx.font = '10px "JetBrains Mono", monospace';
476 |             ctx.textAlign = 'left';
477 |             ctx.fillText(systemState === 'EMERGENCY' ? '■ ALERT: TARGET 01' : 'TRACKING: PATROL-V3', tx - 8, ty - 22);
478 |         }
479 |
480 |         // Crosshairs
481 |         if (showCrosshairs) {
482 |             ctx.strokeStyle = 'rgba(6, 182, 212, 0.45)';
483 |             ctx.lineWidth = 1;
484 |             ctx.beginPath();
485 |             ctx.moveTo(w / 2 - 20, h / 2); ctx.lineTo(w / 2 + 20, h / 2);
486 |             ctx.moveTo(w / 2, h / 2 - 20); ctx.lineTo(w / 2, h / 2 + 20);
487 |             ctx.stroke();
488 |         }
489 |
490 |         // Noise on Congestion or Poor Signal
491 |         if (networkLoad === 'CONGESTED' || signalDbm < -95) {
492 |             for (let i = 0; i < 15; i++) {
493 |                 ctx.fillStyle = Math.random() > 0.5 ? 'rgba(6, 182, 212, 0.3)' : 'rgba(239, 68, 68, 0.3)';
494 |                 ctx.fillRect(Math.random() * w, Math.random() * h, 40, 8);
495 |             }
496 |         }

```

```

497 |         anim = requestAnimationFrame(render);
498 |     };
499 |     anim = requestAnimationFrame(render);
500 |     return () => cancelAnimationFrame(anim);
501 | }, [isTransmitting, systemState, visionMode, showCrosshairs, showMotionBox, networkLoad, signalDbm]);
502 |
503 |
504 | // Telemetry update interval
505 | useEffect(() => {
506 |     const interval = setInterval(() => {
507 |         let lat = 140 + (networkLoad === 'CONGESTED' ? 180 : networkLoad === 'HIGH' ? 80 : 0) + (signalDbm < -95 ? 60 : 0);
508 |         let loss = networkLoad === 'CONGESTED' ? 5.8 : networkLoad === 'HIGH' ? 2.4 : 0.1;
509 |         setTelemetry(t => ({
510 |             ...t,
511 |             bitrateKbps: isTransmitting ? activeCameras * cameraBitrate : 0,
512 |             latencyMs: Math.round(lat + (Math.random() * 6 - 3)),
513 |             packetLossPct: parseFloat(loss.toFixed(1))
514 |         }));
515 |     }, 1000);
516 |     return () => clearInterval(interval);
517 | }, [isTransmitting, networkLoad, signalDbm, activeCameras, cameraBitrate]);
518 |
519 | // Emergency Trigger
520 | const triggerEmergency = () => {
521 |     setSystemState('EMERGENCY');
522 |     const time = new Date().toString().split(' ')[0];
523 |     setEmergencyAlert({
524 |         msg: 'Critical emergency alarm tripped by mobile unit CAMERA-01 (Patrol Van 3). Priority preemption active.',
525 |         time
526 |     });
527 |     setTelemetry(t => ({ ...t, emergencyCount: t.emergencyCount + 1 }));
528 |     sendPacket('ALARM', true);
529 |     addLog('EMERGENCY', '■ EMERGENCY ALARM: Alarm message received from CAMERA-01 (Priority: HIGH)', 'EMERGENCY');
530 |     addAudit('CAMERA-01', 'Operator-01', 'EMERGENCY ALARM', 'HIGH PRIORITY', 'Tactical alarm broadcast to control room.');
```

```

531 | };
532 |
533 | // Scenario Automation
534 | const runFullScenario = () => {
535 |     setSystemState('REGISTERING');
536 |     setIsScenarioRunning(true);
537 |     setScenarioIndex(0);
538 |     setScenarioProgress(0);
539 |     addLog('SCENARIO', 'Automated 7-stage 3G operational scenario initiated.', 'INFO');
540 |     addAudit('CAMERA-01', 'Operator-01', 'Scenario Run', 'SUCCESS', '7-stage sequence initiated');
```

```

541 | };
542 |
543 | useEffect(() => {
544 |     if (!isScenarioRunning) return;
545 |     const step = SCENARIO_STEPS[scenarioIndex];
546 |     const timer = setTimeout(() => {
547 |         setSystemState(step.state);
548 |         if (step.step === 2) setIsTransmitting(true);
549 |         if (step.step === 3) setIsMobilityRunning(true);
550 |         if (step.step === 5) setNetworkLoad('CONGESTED');
551 |         if (step.step === 6) triggerEmergency();
552 |         if (step.step === 7) {
553 |             setIsTransmitting(false);
554 |             setIsMobilityRunning(false);
555 |             setNetworkLoad('LOW');
556 |             setIsScenarioRunning(false);
557 |         }
558 |         if (scenarioIndex < SCENARIO_STEPS.length - 1) {
```

```

559 |         setScenarioIndex(i => i + 1);
560 |     } else {
561 |         setIsScenarioRunning(false);
562 |         addLog('SCENARIO', '7-stage 3G operational scenario completed successfully.', 'SUCCESS');
563 |     }
564 |     }, 4000 / scenarioSpeed);
565 |     return () => clearTimeout(timer);
566 | }, [isScenarioRunning, scenarioIndex, scenarioSpeed]);
567 |
568 | const resetSystem = () => {
569 |     setSystemState('IDLE');
570 |     setIsTransmitting(false);
571 |     setIsMobilityRunning(false);
572 |     setPackets([]);
573 |     setNetworkLoad('LOW');
574 |     setSignalDbm(-80);
575 |     setActiveCameras(1);
576 |     setEmergencyAlert(null);
577 |     setIsScenarioRunning(false);
578 |     setHandoverState({ activeCell: 'Cell A', targetCell: null, status: 'IDLE', cellASignal: -8.2, cellBSignal: -17.5, cellCSignal: -18.5 });
579 |     addLog('SYSTEM', 'System reset to initial state. Ready for new demonstration.', 'INFO');
580 | };
581 |
582 | return (
583 |     <div className="min-h-screen bg-slate-950 text-slate-100 flex flex-col selection:bg-cyan-500/30 selection:text-cyan-200 font-sans">
584 |         { /* Header */ }
585 |         <header className="border-b border-slate-800 bg-slate-950/90 sticky top-0 z-40 px-4 lg:px-8 py-3.5 backdrop-blur shadow-xl">
586 |             <div className="max-w-7xl mx-auto flex flex-col lg:flex-row items-start lg:items-center justify-between gap-4">
587 |                 <div className="flex items-center gap-3">
588 |                     <div className="p-2.5 rounded-xl bg-cyan-950 border border-cyan-800 text-cyan-400">
589 |                         <Icon name="radio" className="w-6 h-6 animate-pulse" />
590 |                     </div>
591 |                     <div>
592 |                         <div className="flex items-center gap-2 flex-wrap">
593 |                             <h1 className="text-lg font-bold text-white font-['Outfit'] m-0">3G Real-Time Communication System</h1>
594 |                             <span className="text-[10px] uppercase font-mono font-semibold px-2 py-0.5 rounded bg-cyan-950 text-cyan-400 border border-cyan-800">
595 |                                 Simulation / Educational Model
596 |                             </span>
597 |                         </div>
598 |                         <p className="text-xs text-slate-400 m-0">Real-Time Mobile Video Surveillance & Emergency Monitoring • UMTS WCDMA
599 |                     </div>
600 |                 </div>
601 |
602 |                 <div className="flex items-center flex-wrap gap-2">
603 |                     <div className={`px-3 py-1.5 rounded-lg border text-xs font-mono font-bold ${
604 |                         systemState === 'STREAMING' ? 'bg-cyan-950 text-cyan-300 border-cyan-500' :
605 |                         systemState === 'EMERGENCY' ? 'bg-red-950 text-red-300 border-red-500 glow-red' :
606 |                         systemState === 'CONGESTED' ? 'bg-amber-950 text-amber-300 border-amber-600' :
607 |                         'bg-slate-800 text-slate-300 border-slate-700'
608 |                     }`}>
609 |                         ● {systemState}
610 |                     </div>
611 |
612 |                     <button
613 |                         onClick={() => setIsTransmitting(!isTransmitting)}
614 |                         className={`px-3.5 py-1.5 rounded-lg font-medium text-xs flex items-center gap-1.5 transition-all cursor-pointer ${
615 |                             isTransmitting ? 'bg-amber-600/25 text-amber-300 border border-amber-500/50' : 'bg-cyan-600 hover:bg-cyan-500 text-cyan-300'
616 |                         }`}
617 |                     >
618 |                         <Icon name={isTransmitting ? "pause" : "play"} className="w-3.5 h-3.5" />
619 |                         <span>{isTransmitting ? 'Pause Stream' : 'Start Live Transmission'}</span>
620 |                     </button>

```

```

621 |
622 |         <button
623 |             onClick={runFullScenario}
624 |             disabled={isScenarioRunning}
625 |             className="px-3 py-1.5 rounded-lg font-medium text-xs bg-indigo-600/20 hover:bg-indigo-600/30 text-indigo-300 border
626 |         >
627 |             <Icon name="forward" className="w-3.5 h-3.5" />
628 |             <span>Run Full Scenario</span>
629 |         </button>
630 |
631 |         <button
632 |             onClick={() => setIsMobilityRunning(!isMobilityRunning)}
633 |             className={`px-3 py-1.5 rounded-lg font-medium text-xs flex items-center gap-1.5 border cursor-pointer ${
634 |                 isMobilityRunning ? 'bg-blue-600/25 text-blue-300 border-blue-500' : 'bg-slate-800 text-slate-300 border-slate-7
635 |             }`}
636 |         >
637 |             <Icon name="car" className="w-3.5 h-3.5" />
638 |             <span>{isMobilityRunning ? 'Patrol Active' : 'Start Mobility'}</span>
639 |         </button>
640 |
641 |         <button
642 |             onClick={triggerEmergency}
643 |             className="px-3.5 py-1.5 rounded-lg font-bold text-xs bg-red-600 hover:bg-red-500 text-white flex items-center gap-1
644 |         >
645 |             <Icon name="alarm" className="w-3.5 h-3.5" />
646 |             <span>Trigger Emergency Alarm</span>
647 |         </button>
648 |
649 |         <button
650 |             onClick={resetSystem}
651 |             className="p-1.5 rounded-lg bg-slate-900 hover:bg-slate-800 text-slate-400 hover:text-white border border-slate-800
652 |             title="Reset System"
653 |         >
654 |             <Icon name="reset" className="w-4 h-4" />
655 |         </button>
656 |     </div>
657 | </div>
658 | </header>
659 |
660 | {/* Navigation Tabs */}
661 | <nav className="bg-slate-900/80 border-b border-slate-800 px-4 lg:px-8 py-2 sticky top-[65px] z-30 backdrop-blur overflow-x-auto" >
662 |     <div className="max-w-7xl mx-auto flex items-center gap-1.5">
663 |         {[
664 |             { id: 'dashboard', label: 'Live Surveillance', icon: 'video', badge: telemetry.bitrateKbps > 0 ? `${telemetry.bitrateKbps} Kbps` : '' },
665 |             { id: 'architecture', label: '3G Architecture & Packets', icon: 'network', badge: `${telemetry.packetsSent} pkts` },
666 |             { id: 'qos', label: 'QoS & Congestion', icon: 'sliders', badge: admissionStatus === 'REJECTED' ? 'REJECTED' : null },
667 |             { id: 'mobility', label: 'Mobility & Handover', icon: 'car', badge: handoverState.activeCell },
668 |             { id: 'calculator', label: 'Capacity Calculator', icon: 'calc', badge: '5.12 Mbps Model' },
669 |             { id: 'security', label: 'Security & 3GPP AKA', icon: 'shield', badge: 'AKA Secure' },
670 |             { id: 'timeline', label: 'Operation Scenario', icon: 'clock', badge: '7 Stages' },
671 |             { id: 'analytics', label: 'Telemetry & Charts', icon: 'chart', badge: `${telemetry.latencyMs} ms` },
672 |             { id: 'reference', label: 'Specs & 3G Guide', icon: 'book', badge: 'Standards' }
673 |         ]}.map(tab => (
674 |             <button
675 |                 key={tab.id}
676 |                 onClick={() => setActiveTab(tab.id)}
677 |                 className={`flex items-center gap-2 px-3 py-1.5 rounded-lg text-xs font-medium whitespace-nowrap transition-all duration-200
678 |                     ${activeTab === tab.id ? 'bg-cyan-500/15 text-cyan-300 border border-cyan-500/40 shadow-sm' : 'text-slate-400 hover:text-slate-200 hover:bg-slate-800/60 border border-transparent'}
679 |             >
680 |                 {tab.icon}
681 |                 {tab.label}
682 |             </button>

```

```

683 |         <Icon name={tab.icon} className="w-3.5 h-3.5" />
684 |         <span>{tab.label}</span>
685 |         {tab.badge && (
686 |           <span className={`text-[10px] font-mono px-1.5 py-0.2 rounded ${
687 |             tab.badge === 'REJECTED' ? 'bg-red-950 text-red-400 border border-red-800' : 'bg-cyan-950 text-cyan-400 border
688 |           `}>
689 |             {tab.badge}
690 |           </span>
691 |         )}
692 |       </button>
693 |     )}
694 |   </div>
695 | </nav>
696 |
697 |   {/} Main Views Container */}
698 |   <main className="flex-1 max-w-7xl w-full mx-auto p-4 lg:p-8 space-y-6">
699 |     {/} View 1: Dashboard / Live Surveillance */}
700 |     {activeTab === 'dashboard' && (
701 |       <div className="space-y-6">
702 |         {emergencyAlert && (
703 |           <div className="rounded-xl border border-red-500 bg-red-950/80 p-3.5 flex items-center justify-between text-red-200">
704 |             <div className="flex items-center gap-3">
705 |               <Icon name="alarm" className="w-5 h-5 text-red-400 animate-bounce" />
706 |               <div>
707 |                 <strong className="text-white font-mono text-sm block">■ EMERGENCY ALERT ENGAGED</strong>
708 |                 <span className="text-xs text-red-300">{emergencyAlert.msg}</span>
709 |               </div>
710 |             </div>
711 |             <button onClick={() => setEmergencyAlert(null)} className="px-3 py-1 rounded bg-red-800 hover:bg-red-700 text-white">
712 |               Acknowledge
713 |             </button>
714 |           </div>
715 |         )}
716 |
717 |         <div className="grid grid-cols-1 lg:grid-cols-12 gap-6">
718 |           {/} Surveillance Feed */}
719 |           <div className="lg:col-span-8 space-y-3">
720 |             <div className="rounded-2xl border border-slate-800 bg-slate-900/90 overflow-hidden shadow-2xl relative">
721 |               <div className="bg-slate-950/90 border-b border-slate-800 px-4 py-2.5 flex items-center justify-between text-white">
722 |                 <div className="flex items-center gap-3">
723 |                   <span className={`w-2.5 h-2.5 rounded-full ${isTransmitting ? 'bg-red-500 animate-ping' : 'bg-slate-600'}>
724 |                     <span className={`font-bold ${isTransmitting ? 'text-red-400' : 'text-slate-500'}>{isTransmitting ? 'LIVE' : 'OFFLINE'}</span>
725 |                   <span className="text-slate-600">|</span>
726 |                   <span className="text-cyan-400 font-bold">{activeCamId}</span>
727 |                   <span className="text-slate-400 hidden sm:inline">(Mobile Patrol Unit Alpha)</span>
728 |                 </div>
729 |                 <div className="text-cyan-300 font-bold">{telemetry.bitrateKbps} kbps • H.264 Baseline</div>
730 |               </div>
731 |
732 |               <div className="relative aspect-video bg-slate-950 flex items-center justify-center overflow-hidden">
733 |                 <canvas ref={canvasRef} width={640} height={360} className="w-full h-full object-cover" />
734 |                 <div className="absolute inset-0 surveillance-scanlines opacity-50 pointer-events-none" />
735 |               </div>
736 |
737 |               <div className="bg-slate-950/90 border-t border-slate-800 px-4 py-3 flex flex-wrap items-center justify-between">
738 |                 <div className="flex items-center gap-2">
739 |                   {[ 'CAM-01', 'CAM-02', 'CAM-03' ].map(cam => (
740 |                     <button
741 |                       key={cam}
742 |                       onClick={() => setActiveCamId(cam)}
743 |                       className={`px-2.5 py-1 rounded font-mono font-bold cursor-pointer ${activeCamId === cam ? 'bg-cyan-600' : 'bg-slate-800'}>
744 |                     >

```

```

745 |         {cam}
746 |     </button>
747 |     )})
748 | </div>
749 |
750 |     <div className="flex items-center gap-2">
751 |         <button onClick={() => setVisionMode('NORMAL')} className={`px-2.5 py-1 rounded cursor-pointer ${visionMode === 'NORMAL' ? 'bg-emerald-500' : 'bg-slate-400'} text-white`}>
752 |             Day/Night
753 |         </button>
754 |         <button onClick={() => setVisionMode('IR_THERMAL')} className={`px-2.5 py-1 rounded cursor-pointer ${visionMode === 'IR_THERMAL' ? 'bg-emerald-500' : 'bg-slate-400'} text-white`}>
755 |             Thermal IR
756 |         </button>
757 |         <button onClick={() => setShowCrosshairs(!showCrosshairs)} className={`p-1.5 rounded cursor-pointer ${showCrosshairs ? 'bg-emerald-500' : 'bg-slate-400'} text-white`}>
758 |             <Icon name="crosshair" className="w-3.5 h-3.5" />
759 |         </button>
760 |         <button onClick={() => setShowMotionBox(!showMotionBox)} className={`p-1.5 rounded cursor-pointer ${showMotionBox ? 'bg-emerald-500' : 'bg-slate-400'} text-white`}>
761 |             <Icon name="eye" className="w-3.5 h-3.5" />
762 |         </button>
763 |     </div>
764 | </div>
765 | </div>
766 | </div>
767 |
768 | {/* Right: System Information Panel */}
769 | <div className="lg:col-span-4 space-y-4">
770 |     <div className="rounded-2xl border border-slate-800 bg-slate-900/90 p-5 shadow-xl space-y-4">
771 |         <div className="flex items-center justify-between border-b border-slate-800 pb-3">
772 |             <h3 className="text-sm font-bold uppercase tracking-wider text-slate-200 font-mono m-0">System Information Panel</h3>
773 |             <span className="text-[10px] font-mono px-2 py-0.5 rounded bg-emerald-950 text-emerald-400 border border-emerald-900">View 1</span>
774 |         </div>
775 |
776 |         <div className="divide-y divide-slate-800/80 font-mono text-xs">
777 |             <div className="py-2 flex justify-between"><span className="text-slate-400">Connection:</span><span className="text-slate-400">Technology:</span><span className="text-slate-400">Signal:</span></div>
778 |             <div className="py-2 flex justify-between"><span className="text-slate-400">Technology:</span><span className="text-slate-400">Bitrate:</span><span className="text-slate-400">Packet Loss:</span></div>
779 |             <div className="py-2 flex justify-between"><span className="text-slate-400">Bitrate:</span><span className="text-slate-400">Latency:</span><span className="text-slate-400">QoS:</span></div>
780 |             <div className="py-2 flex justify-between"><span className="text-slate-400">Latency:</span><span className="text-slate-400">Packet Loss:</span><span className="text-slate-400">Authentication:</span></div>
781 |             <div className="py-2 flex justify-between"><span className="text-slate-400">Packet Loss:</span><span className="text-slate-400">Signal:</span><span className="text-slate-400">Session:</span></div>
782 |             <div className="py-2 flex justify-between"><span className="text-slate-400">Signal:</span><span className="text-slate-400">QoS:</span><span className="text-slate-400">End-to-End Delay:</span></div>
783 |             <div className="py-2 flex justify-between"><span className="text-slate-400">QoS:</span><span className="text-slate-400">Authentication:</span><span className="text-slate-400">End-to-End Delay:</span></div>
784 |             <div className="py-2 flex justify-between"><span className="text-slate-400">Authentication:</span><span className="text-slate-400">Session:</span><span className="text-slate-400">End-to-End Delay:</span></div>
785 |             <div className="py-2 flex justify-between"><span className="text-slate-400">Session:</span><span className="text-slate-400">End-to-End Delay:</span><span className="text-slate-400">End-to-End Delay:</span></div>
786 |         </div>
787 |
788 |         <button onClick={triggerEmergency} className="w-full py-2.5 rounded-xl bg-red-600 hover:bg-red-500 text-white">
789 |             <Icon name="alarm" className="w-4 h-4" />
790 |             Trigger Mobile Emergency Alarm</button>
791 |     </div>
792 | </div>
793 | </div>
794 | </div>
795 | </div>
796 | })
797 |
798 | {/* View 2: Architecture & Packets */}
799 | {activeTab === 'architecture' && (
800 |     <div className="space-y-6">
801 |         <div className="rounded-2xl border border-slate-800 bg-slate-900/90 p-5 shadow-xl">
802 |             <div className="flex flex-col sm:flex-row items-start sm:items-center justify-between gap-3 border-b border-slate-800">
803 |                 <div>
804 |                     <h3 className="text-base font-bold text-white font-['Outfit'] m-0">End-to-End 3G Communication Architecture</h3>
805 |                     <p className="text-xs text-slate-400 m-0">Click any node to inspect internal hardware, protocols, and telecommunication details.</p>
806 |                 </div>

```

```

807 |         <div className="flex items-center gap-1.5">
808 |             <button onClick={() => sendPacket('VIDEO')} className="px-2.5 py-1 rounded bg-cyan-950 text-cyan-300 border bo
809 |             <button onClick={() => sendPacket('AUDIO')} className="px-2.5 py-1 rounded bg-purple-950 text-purple-300 borde
810 |             <button onClick={() => sendPacket('CONTROL')} className="px-2.5 py-1 rounded bg-amber-950 text-amber-300 borde
811 |             <button onClick={() => sendPacket('BACKGROUND')} className="px-2.5 py-1 rounded bg-slate-800 text-slate-300 bo
812 |         </div>
813 |     </div>
814 |
815 |     {/ * 7 Nodes Topology */}
816 |     <div className="overflow-x-auto pb-6 pt-2">
817 |         <div className="min-w-[1020px] flex items-center justify-between gap-2 px-2">
818 |             {SYSTEM_NODES.map((n, i) => (
819 |                 <React.Fragment key={n.id}>
820 |                     <div
821 |                         onClick={() => setSelectedNode(n)}
822 |                         className={`w-36 rounded-xl p-3.5 border transition-all cursor-pointer select-none ${
823 |                             activeNodePulse === n.id ? 'border-cyan-400 bg-cyan-950/70 scale-105 shadow-[0_0_20px_rgba(6,182,212,0
824 |                             systemState === 'EMERGENCY' ? 'border-red-500 bg-red-950/50' : 'border-slate-800 bg-slate-950 hover:bo
825 |                         ]`}
826 |                     >
827 |                         <span className="text-[9px] font-mono text-cyan-400 block font-bold uppercase">{n.category}</span>
828 |                         <h4 className="text-xs font-bold text-white font-['Outfit'] mt-1 m-0">{n.shortName}</h4>
829 |                         <p className="text-[10px] text-slate-400 font-mono mt-1 line-clamp-2 leading-tight">{n.subtitle}</p>
830 |                     </div>
831 |
832 |                     {i < SYSTEM_NODES.length - 1 && (
833 |                         <div className="relative flex-1 min-w-[50px] max-w-[90px] h-14 flex flex-col items-center justify-center
834 |                             <span className="text-[9px] font-mono text-slate-500 mb-1">
835 |                                 {[ 'Uu Air', 'Iub', 'Iu-PS', 'Gn/Gi', 'IPsec', 'RTSP/RTP' ][i]}
836 |                             </span>
837 |                             <div className="w-full h-1 bg-slate-800 rounded relative overflow-hidden" />
838 |                             <div className="absolute inset-0 pointer-events-none flex items-center">
839 |                                 {packets.filter(p => p.currentLink === i).map(p => (
840 |                                     <div key={p.id} className="absolute transform -translate-y-1/2" style={{ left: `${Math.min(92, Mat
841 |                                     <span className={`text-[8px] font-mono font-bold px-1.5 py-0.5 rounded shadow ${p.isAlarm ? 'bg-
842 |                                     {p.label}
843 |                                     </span>
844 |                                 </div>
845 |                             )))
846 |                             </div>
847 |                         </div>
848 |                     ))}
849 |                 </React.Fragment>
850 |             ))}
851 |         </div>
852 |     </div>
853 | </div>
854 | </div>
855 | ))}
856 |
857 | {/ * View 3: QoS & Congestion */}
858 | {activeTab === 'qos' && (
859 |     <div className="space-y-6">
860 |         {networkLoad === 'CONGESTED' && (
861 |             <div className="rounded-xl border border-amber-500/80 bg-amber-950/70 p-4 text-amber-200 text-xs">
862 |                 <strong className="text-amber-100 font-bold block mb-1">QoS / Admission Mechanisms Protecting Priority Traffic</
863 |                 <span>Network is CONGESTED. Low-priority background telemetry is throttled so real-time video (256 kbps) passes
864 |             </div>
865 |         )}
866 |         {admissionStatus === 'REJECTED' && (
867 |             <div className="rounded-xl border border-red-500/80 bg-red-950/80 p-4 text-red-200 text-xs glow-red">
868 |                 <strong className="text-white font-bold block mb-1">Admission Control: New Session Rejected Due to Insufficient

```

```

869 |         <span>Exceeded 3,840 kbps cell threshold. New camera sessions blocked to preserve quality for active streams.</span>
870 |     </div>
871 | })
872 |
873 | <div className="grid grid-cols-1 lg:grid-cols-12 gap-6">
874 |     <div className="lg:col-span-7 rounded-2xl border border-slate-800 bg-slate-900/90 p-5 shadow-xl space-y-4">
875 |         <h3 className="text-sm font-bold uppercase tracking-wider text-slate-200 font-mono m-0">UMTS Traffic Prioritization</h3>
876 |         <table className="w-full text-left text-xs font-mono">
877 |             <thead>
878 |                 <tr>
879 |                     <th className="border-b border-slate-800 text-slate-400">
880 |                         <div>
881 |                             <div>
882 |                                 <div>
883 |                                     <div>
884 |                                     </div>
885 |                                 </div>
886 |                             </div>
887 |                         </th>
888 |                     <th className="border-b border-slate-800 text-slate-400">
889 |                         <div>
890 |                             <div>
891 |                                 <div>
892 |                                     <div>
893 |                                     </div>
894 |                                 </div>
895 |                             </div>
896 |                         </th>
897 |                     <th className="border-b border-slate-800 text-slate-400">
898 |                         <div>
899 |                             <div>
900 |                                 <div>
901 |                                     <div>
902 |                                     </div>
903 |                                 </div>
904 |                             </div>
905 |                         </th>
906 |                     <th className="border-b border-slate-800 text-slate-400">
907 |                         <div>
908 |                             <div>
909 |                                 <div>
910 |                                     <div>
911 |                                     </div>
912 |                                 </div>
913 |                             </div>
914 |                         </th>
915 |                     <th className="border-b border-slate-800 text-slate-400">
916 |                         <div>
917 |                             <div>
918 |                                 <div>
919 |                                     <div>
920 |                                     </div>
921 |                                 </div>
922 |                             </div>
923 |                         </th>
924 |                     <th className="border-b border-slate-800 text-slate-400">
925 |                         <div>
926 |                             <div>
927 |                                 <div>
928 |                                     <div>
929 |                                     </div>
930 |                                 </div>
931 |                             </div>
932 |                         </th>
933 |                     <th className="border-b border-slate-800 text-slate-400">
934 |                         <div>
935 |                             <div>
936 |                                 <div>
937 |                                     <div>
938 |                                     </div>
939 |                                 </div>
940 |                             </div>
941 |                         </th>
942 |                     <th className="border-b border-slate-800 text-slate-400">
943 |                         <div>
944 |                             <div>
945 |                                 <div>
946 |                                     <div>
947 |                                     </div>
948 |                                 </div>
949 |                             </div>
950 |                         </th>
951 |                     <th className="border-b border-slate-800 text-slate-400">
952 |                         <div>
953 |                             <div>
954 |                                 <div>
955 |                                     <div>
956 |                                     </div>
957 |                                 </div>
958 |                             </div>
959 |                         </th>
960 |                     <th className="border-b border-slate-800 text-slate-400">
961 |                         <div>
962 |                             <div>
963 |                                 <div>
964 |                                     <div>
965 |                                     </div>
966 |                                 </div>
967 |                             </div>
968 |                         </th>
969 |                     <th className="border-b border-slate-800 text-slate-400">
970 |                         <div>
971 |                             <div>
972 |                                 <div>
973 |                                     <div>
974 |                                     </div>
975 |                                 </div>
976 |                             </div>
977 |                         </th>
978 |                     <th className="border-b border-slate-800 text-slate-400">
979 |                         <div>
980 |                             <div>
981 |                                 <div>
982 |                                     <div>
983 |                                     </div>
984 |                                 </div>
985 |                             </div>
986 |                         </th>
987 |                     <th className="border-b border-slate-800 text-slate-400">
988 |                         <div>
989 |                             <div>
990 |                                 <div>
991 |                                     <div>
992 |                                     </div>
993 |                                 </div>
994 |                             </div>
995 |                         </th>
996 |                     <th className="border-b border-slate-800 text-slate-400">
997 |                         <div>
998 |                             <div>
999 |                                 <div>
1000 |                                     <div>

```

```

931 |         </div>
932 |
933 |         <div className="relative w-full h-64 bg-slate-950 rounded-xl border border-slate-800 overflow-hidden telecom-grid"
934 |             {[ 'Cell A', 'Cell B', 'Cell C' ].map((c, idx) => (
935 |                 <div key={c} className={`w-44 h-44 sm:w-52 sm:h-52 rounded-full border-2 border-dashed flex flex-col items-center
936 |                     handoverState.activeCell === c ? 'border-cyan-400 bg-cyan-950/25 shadow-[0_0_25px_rgba(6,182,212,0.3)]' : 'border-slate-800 bg-slate-950'`}>
937 |                 </div>
938 |                 <span className="text-sm font-bold text-white font-mono">{c}</span>
939 |                 <span className="text-[10px] text-slate-400 font-mono">Node B {idx + 1}</span>
940 |                 <span className="text-xs font-mono text-cyan-300 font-bold mt-1">
941 |                     {[handoverState.cellASignal, handoverState.cellBSignal, handoverState.cellCSignal][idx]} dB Ec/No
942 |                 </span>
943 |             </div>
944 |         ))}
945 |
946 |         <div className="absolute z-20 flex flex-col items-center transition-all duration-300" style={{ left: `${Math.min(handoverState.distanceKm, 100)}px`}}>
947 |             <div className="p-2 rounded-xl bg-cyan-500 text-slate-950 font-bold shadow-lg animate-pulse">
948 |                 <Icon name="car" className="w-5 h-5" />
949 |             </div>
950 |             <span className="text-[10px] font-mono font-bold text-white bg-slate-900 px-2 py-0.5 rounded border border-cyan-500">
951 |                 CAMERA-01 ({handoverState.distanceKm} km)
952 |             </span>
953 |         </div>
954 |     </div>
955 |
956 |     <div className="grid grid-cols-2 sm:grid-cols-4 gap-3 font-mono text-xs">
957 |         <div className="p-3 rounded-xl bg-slate-950 border border-slate-800"><span className="text-slate-500 block text-ellipsis">Node A</span></div>
958 |         <div className="p-3 rounded-xl bg-slate-950 border border-slate-800"><span className="text-slate-500 block text-ellipsis">Node B</span></div>
959 |         <div className="p-3 rounded-xl bg-slate-950 border border-slate-800"><span className="text-slate-500 block text-ellipsis">Node C</span></div>
960 |         <div className="p-3 rounded-xl bg-slate-950 border border-slate-800"><span className="text-slate-500 block text-ellipsis">Node D</span></div>
961 |     </div>
962 | </div>
963 | </div>
964 | ))}
965 |
966 | /* View 5: Capacity Calculator */
967 | {activeTab === 'calculator' && (
968 |     <div className="space-y-6">
969 |         <div className="grid grid-cols-1 lg:grid-cols-12 gap-6">
970 |             <div className="lg:col-span-6 rounded-2xl border border-slate-800 bg-slate-900/90 p-6 shadow-xl space-y-4">
971 |                 <h3 className="text-sm font-bold uppercase tracking-wider text-slate-200 font-mono m-0">Capacity Dimensioning Parameters</h3>
972 |                 <div className="space-y-2">
973 |                     <div className="flex justify-between text-xs font-mono"><span className="text-slate-300">Number of Simultaneous Calls</span></div>
974 |                     <input type="range" min="1" max="50" value={activeCameras} onChange={e => setActiveCameras(parseInt(e.target.value))}>
975 |                 </div>
976 |                 <div className="space-y-2">
977 |                     <div className="flex justify-between text-xs font-mono"><span className="text-slate-300">Bitrate per Camera</span></div>
978 |                     <div className="grid grid-cols-4 gap-2">
979 |                         {[64, 128, 256, 384].map(r => (
980 |                             <button key={r} onClick={() => setCameraBitrate(r)} className={`py-1.5 rounded text-xs font-mono font-bold ${activeCameraBitrate === r ? 'bg-cyan-500 text-slate-950' : 'bg-slate-800 text-slate-300'}>{r}</button>
981 |                         ))}
982 |                     </div>
983 |                 </div>
984 |             </div>
985 |
986 |             <div className="lg:col-span-6 rounded-2xl border border-cyan-500/40 bg-slate-900/90 p-6 shadow-xl space-y-4">
987 |                 <h3 className="text-sm font-bold uppercase tracking-wider text-cyan-400 font-mono m-0">Aggregated Traffic Calculation</h3>
988 |                 <div className="p-4 rounded-xl bg-slate-950 border border-slate-800 text-center">
989 |                     <span className="text-xs font-mono uppercase text-slate-400 block mb-1">Total Application Traffic</span>
990 |                     <div className="text-4xl font-extrabold text-cyan-400 font-['Outfit']">
991 |                         {((activeCameras * cameraBitrate) / 1000).toFixed(2)} <span className="text-xl text-slate-300">Mbps</span>
992 |                     </div>

```

```

993 |         <span className="text-xs font-mono text-slate-400 block mt-1">({activeCameras * cameraBitrate} kbps payload su
994 |     </div>
995 |
996 |     <div className="p-3 rounded-xl bg-emerald-950/40 border border-emerald-800 text-xs font-mono text-emerald-300">
997 |         <strong>Assignment Reference Case:</strong> 20 cameras × 256 kbps = <strong>5.12 Mbps</strong>
998 |     </div>
999 |
1000 |     <div className="p-3 rounded-xl bg-slate-950 border border-slate-800 text-[11px] text-slate-400 space-y-1">
1001 |         <p className="m-0">"This calculation is before protocol overhead and network engineering margins."</p>
1002 |         <p className="m-0">"The actual number of supported users depends on cell loading, radio conditions, bearer con
1003 |     </div>
1004 | </div>
1005 | </div>
1006 | </div>
1007 | })
1008 |
1009 | /* View 6: Security & 3GPP AKA */
1010 | {activeTab === 'security' && (
1011 |     <div className="space-y-6">
1012 |         <div className="grid grid-cols-2 sm:grid-cols-5 gap-3 font-mono text-xs">
1013 |             <div className="p-3 rounded-xl bg-slate-900 border border-slate-800"><span className="text-slate-500 block text-[1
1014 |             <div className="p-3 rounded-xl bg-slate-900 border border-slate-800"><span className="text-slate-500 block text-[1
1015 |             <div className="p-3 rounded-xl bg-slate-900 border border-slate-800"><span className="text-slate-500 block text-[1
1016 |             <div className="p-3 rounded-xl bg-slate-900 border border-slate-800"><span className="text-slate-500 block text-[1
1017 |             <div className="p-3 rounded-xl bg-slate-900 border border-slate-800"><span className="text-slate-500 block text-[1
1018 |         </div>
1019 |
1020 |         <div className="rounded-2xl border border-slate-800 bg-slate-900/90 p-5 shadow-xl space-y-3 font-mono text-xs">
1021 |             <h3 className="text-sm font-bold uppercase text-slate-200 m-0">Security & Operational Audit Log</h3>
1022 |             <table className="w-full text-left">
1023 |                 <thead>
1024 |                     <tr className="border-b border-slate-800 text-slate-400">
1025 |                         <th className="pb-2">Time</th><th className="pb-2">Device</th><th className="pb-2">User</th><th className="pb-2">P
1026 |                     </tr>
1027 |                 </thead>
1028 |                 <tbody className="divide-y divide-slate-800/80">
1029 |                     {auditLogs.map(l => (
1030 |                         <tr key={l.id}>
1031 |                             <td className="py-2 text-slate-400">{l.time}</td>
1032 |                             <td className="py-2 text-cyan-300 font-bold">{l.device}</td>
1033 |                             <td className="py-2 text-slate-300">{l.user}</td>
1034 |                             <td className="py-2 text-white font-medium">{l.action}</td>
1035 |                             <td className="py-2"><span className={`px-2 py-0.5 rounded text-[10px] font-bold ${l.status === 'SUCCESS'
1036 |                             <td className="py-2 text-slate-400">{l.details}</td>
1037 |                         </tr>
1038 |                     )))
1039 |                 </tbody>
1040 |             </table>
1041 |         </div>
1042 |     </div>
1043 | })
1044 |
1045 | /* View 7: Operation Scenario */
1046 | {activeTab === 'timeline' && (
1047 |     <div className="space-y-4">
1048 |         <div className="rounded-2xl border border-indigo-500/40 bg-slate-900/90 p-5 shadow-xl flex flex-col sm:flex-row item
1049 |             <div>
1050 |                 <h3 className="text-base font-bold text-white font-['Outfit'] m-0">Real-Time Operation Sequence (7 Stages)</h3>
1051 |                 <p className="text-xs text-slate-400 m-0">From camera power-up to mission debrief.</p>
1052 |             </div>
1053 |             <button onClick={runFullScenario} disabled={isScenarioRunning} className="px-4 py-2 rounded-xl bg-indigo-600 hover:
1054 |                 <Icon name="play" className="w-3.5 h-3.5" />

```

```

1055 |         <span>RUN FULL SCENARIO</span>
1056 |     </button>
1057 | </div>
1058 |
1059 | <div className="space-y-2.5">
1060 |     {SCENARIO_STEPS.map((s, idx) => (
1061 |         <div key={s.step} className={`rounded-xl border p-4 transition-all ${
1062 |             isScenarioRunning && scenarioIndex === idx ? 'border-indigo-500 bg-indigo-950/40 shadow-lg' : 'border-slate-800
1063 |         }`}>
1064 |             <div className="flex items-center justify-between mb-1">
1065 |                 <span className="text-xs font-mono font-bold text-indigo-400">STAGE {s.step}: {s.title}</span>
1066 |                 <span className="text-[10px] font-mono text-cyan-300">{s.state}</span>
1067 |             </div>
1068 |             <p className="text-xs text-slate-300 leading-relaxed m-0">{s.desc}</p>
1069 |         </div>
1070 |     ))}
1071 | </div>
1072 | </div>
1073 | ))}
1074 |
1075 | {/* View 8: Telemetry & Event Log */}
1076 | {activeTab === 'analytics' && (
1077 |     <div className="space-y-6">
1078 |         <div className="grid grid-cols-2 sm:grid-cols-4 gap-3 font-mono">
1079 |             <div className="p-4 rounded-xl bg-slate-900 border border-slate-800"><span className="text-[10px] text-slate-500 b
1080 |             <div className="p-4 rounded-xl bg-slate-900 border border-slate-800"><span className="text-[10px] text-slate-500 b
1081 |             <div className="p-4 rounded-xl bg-slate-900 border border-slate-800"><span className="text-[10px] text-slate-500 b
1082 |             <div className="p-4 rounded-xl bg-slate-900 border border-slate-800"><span className="text-[10px] text-slate-500 b
1083 |         </div>
1084 |
1085 |         {/* Event Log Component */}
1086 |         <div className="rounded-2xl border border-slate-800 bg-slate-900/90 p-5 shadow-xl space-y-3 font-mono">
1087 |             <div className="flex justify-between items-center border-b border-slate-800 pb-2.5">
1088 |                 <h3 className="text-xs font-bold uppercase text-slate-200 m-0">Real-Time Event Log ({eventLogs.length} Events)</h3>
1089 |                 <button onClick={() => setEventLogs([])} className="px-2 py-0.5 rounded bg-slate-800 text-slate-400 hover:text-w
1090 |             </div>
1091 |             <div className="h-48 overflow-y-auto space-y-1 text-xs pr-1">
1092 |                 {eventLogs.map(l => (
1093 |                     <div key={l.id} className="py-1 px-2 rounded hover:bg-slate-800/40 flex items-start gap-2.5">
1094 |                         <span className="text-slate-500 shrink-0 text-[11px]">[{l.time}]</span>
1095 |                         <span className={`px-1.5 py-0.2 rounded border text-[9px] uppercase shrink-0 ${l.type === 'EMERGENCY' ? 'tex
1096 |                         <span className="text-slate-300 text-[11px]">{l.msg}</span>
1097 |                     </div>
1098 |                 ))}
1099 |             </div>
1100 |         </div>
1101 |     </div>
1102 | ))}
1103 |
1104 | {/* View 9: Specs & References */}
1105 | {activeTab === 'reference' && (
1106 |     <div className="space-y-6">
1107 |         <div className="rounded-2xl border border-slate-800 bg-slate-900/90 p-5 shadow-xl space-y-4">
1108 |             <h3 className="text-sm font-bold uppercase tracking-wider text-white font-mono m-0">3G Communication System Equipm
1109 |             <table className="w-full text-left text-xs font-mono">
1110 |                 <thead>
1111 |                     <tr className="border-b border-slate-800 text-slate-400">
1112 |                         <th className="pb-2">Component</th><th className="pb-2">Purpose</th>
1113 |                     </tr>
1114 |                 </thead>
1115 |                 <tbody className="divide-y divide-slate-800/80">
1116 |                     <tr><td className="py-2.5 font-bold text-cyan-300">IP camera / mobile camera</td><td className="py-2.5 text-sl

```

```

1117 |         <tr><td className="py-2.5 font-bold text-cyan-300">Video encoder</td><td className="py-2.5 text-slate-300">Com
1118 |         <tr><td className="py-2.5 font-bold text-cyan-300">3G modem / UE</td><td className="py-2.5 text-slate-300">Con
1119 |         <tr><td className="py-2.5 font-bold text-cyan-300">Node B</td><td className="py-2.5 text-slate-300">Provide WC
1120 |         <tr><td className="py-2.5 font-bold text-cyan-300">RNC</td><td className="py-2.5 text-slate-300">Manage radio
1121 |         <tr><td className="py-2.5 font-bold text-cyan-300">3G packet core</td><td className="py-2.5 text-slate-300">Au
1122 |         <tr><td className="py-2.5 font-bold text-cyan-300">Application / streaming server</td><td className="py-2.5 te
1123 |         <tr><td className="py-2.5 font-bold text-cyan-300">Monitoring PC / display</td><td className="py-2.5 text-slat
1124 |     </tbody>
1125 | </table>
1126 | </div>
1127 |
1128 |     <div className="grid grid-cols-1 md:grid-cols-2 gap-4 text-xs font-mono">
1129 |         <div className="p-4 rounded-xl bg-slate-900 border border-slate-800">
1130 |             <span className="text-cyan-400 font-bold block mb-1">3GPP TS 23.060</span>
1131 |             <span className="text-slate-300 block">General Packet Radio Service (GPRS); Service description; Stage 2</span>
1132 |         </div>
1133 |         <div className="p-4 rounded-xl bg-slate-900 border border-slate-800">
1134 |             <span className="text-cyan-400 font-bold block mb-1">3GPP TS 25.401</span>
1135 |             <span className="text-slate-300 block">UTRAN Overall Description (Node B & RNC architecture)</span>
1136 |         </div>
1137 |         <div className="p-4 rounded-xl bg-slate-900 border border-slate-800">
1138 |             <span className="text-cyan-400 font-bold block mb-1">3GPP TS 23.107</span>
1139 |             <span className="text-slate-300 block">QoS Concept and Architecture (Conversational, Streaming, Interactive, Bac
1140 |         </div>
1141 |         <div className="p-4 rounded-xl bg-slate-900 border border-slate-800">
1142 |             <span className="text-cyan-400 font-bold block mb-1">ITU-R Recommendation M.1457</span>
1143 |             <span className="text-slate-300 block">Detailed specifications of the radio interfaces of IMT-2000</span>
1144 |         </div>
1145 |     </div>
1146 | </div>
1147 | })
1148 | </main>
1149 |
1150 | {/* Node Inspector Modal */}
1151 | {selectedNode && (
1152 |     <div className="fixed inset-0 z-50 flex items-center justify-center p-4 bg-slate-950/80 backdrop-blur-sm">
1153 |         <div className="max-w-xl w-full rounded-2xl border border-cyan-500/40 bg-slate-900 p-6 shadow-2xl relative">
1154 |             <button onClick={() => setSelectedNode(null)} className="absolute top-4 right-4 p-1.5 rounded-lg bg-slate-800 text-s
1155 |                 <Icon name="close" className="w-5 h-5" />
1156 |             </button>
1157 |             <span className="text-[10px] font-mono text-cyan-400 font-bold uppercase">{selectedNode.category}</span>
1158 |             <h3 className="text-xl font-bold text-white font-['Outfit'] mt-1 mb-0.5">{selectedNode.name}</h3>
1159 |             <p className="text-xs text-slate-400 font-mono mb-4">{selectedNode.subtitle}</p>
1160 |             <div className="space-y-3 text-xs">
1161 |                 <div className="p-3 rounded-lg bg-slate-950 border border-slate-800"><strong className="text-cyan-300 block mb-1">
1162 |                 <div className="p-3 rounded-lg bg-slate-950 border border-slate-800"><strong className="text-cyan-300 block mb-1">
1163 |                 {selectedNode.subComponents && (
1164 |                     <div className="p-3 rounded-lg bg-slate-950 border border-slate-800"><strong className="text-cyan-300 block mb-1">
1165 |                         <div className="flex flex-wrap gap-1 mt-1">
1166 |                             {selectedNode.subComponents.map((s, i) => (
1167 |                                 <span key={i} className="px-2 py-0.5 rounded bg-slate-800 text-cyan-300 font-mono text-[10px]">{s}</span>
1168 |                             ))}
1169 |                         </div>
1170 |                     </div>
1171 |                 )}
1172 |             </div>
1173 |         </div>
1174 |     </div>
1175 | })
1176 |
1177 | {/* Footer */}
1178 | <footer className="border-t border-slate-900 bg-slate-950/80 py-5 px-4 lg:px-8 mt-12 text-xs font-mono text-slate-500">

```

```
1179 |         <div className="max-w-7xl mx-auto flex flex-col sm:flex-row items-center justify-between gap-4">
1180 |           <span className="text-slate-400 font-semibold">Design of a 3G Communication System for Real-Time Applications</span>
1181 |           <div className="flex items-center gap-3 text-[11px] text-slate-400">
1182 |             <span>UMTS WCDMA FDD (3.84 Mcps)</span>
1183 |             <span>•</span>
1184 |             <span>3GPP Release 99/4/5</span>
1185 |             <span>•</span>
1186 |             <span className="text-cyan-400 font-bold">Academic Simulation Model</span>
1187 |           </div>
1188 |         </div>
1189 |       </footer>
1190 |     </div>
1191 |   );
1192 | }
1193 |
1194 | ReactDOM.createRoot(document.getElementById('root')).render(<App />);
1195 | </script>
1196 | </body>
1197 | </html>
```